Third System
AI Presence Index

AI Availability as a Third Measurable Layer

Five-substrate empirical anchor base under locked methodology v1.6. The Presence component of the AIAS™ construct reaches its 1.0 milestone with cosmetics as the strongest single-phase evidence base: cultural-channel Recall populates the dissociation construct's third quadrant above threshold, Identity Load empirically tracks AI channel asymmetry at confidence-interval rigor, and off-panel brand presence enters as a measured component with channel signatures that track the same Identity-Load gradient as panel-internal brands.

May 2026 · Pablo Ulpiano González Castro · Third System

Independent measurement for the AI mediation layer.

AIAS™ measures AI Availability — the probability that an AI intermediary retrieves, recommends, or selects a brand in a category-anchored decision context — as a third measurable channel of brand presence alongside the Ehrenberg-Bass framework's Mental Availability and Physical Availability. With the v0.21 cosmetics phase, the protocol's empirical anchor base spans five substrate families (kitchenware, indie fragrance, audiophile electronics, skincare, cosmetics) under one locked methodology version. AIAS™ 1.0 anchors the Presence component across five substrate families.

This report consolidates the AIAS™ Measurement Program's five-substrate empirical anchor base in five propositions for brand strategy and marketing-science practitioners. P1: AI Availability is now anchored across five substrate families — kitchenware, indie fragrance, audiophile electronics, skincare, and cosmetics — as a measurable third channel of brand presence. P2: brands not on a measurement panel can still drive AI retrieval, and their channel signature tracks Identity Load — Estée Lauder and Clinique canonical-pure, Glossier cultural-pure. P3: Recall splits into a canonical channel and a cultural channel that can be managed separately, with Rare Beauty's 1:17 R_cat:R_cult split as the textbook diagnostic. P4: Identity Load predicts which channel will lead, with cosmetics returning the program's first CONFIRMED moderator verdict at any layer. P5: methodology v1.6 closes the operational layer that makes P2 through P4 measurable.

AIAS™ measures AI Availability — the probability that an AI intermediary retrieves, recommends, or selects a brand in a category-anchored decision context — as a third measurable channel of brand presence alongside the Ehrenberg-Bass framework's Mental Availability and Physical Availability. The construct is operationalized through a pre-registered measurement protocol that runs against a fixed reference panel of six LLMs and is evaluated under locked verdict matrices each phase. AIAS™ 1.0 names the Presence component of a multi-component AIAS construct; the remaining five components (Ranking, Consistency, Coverage, Grounding, Sentiment) are reserved for a Phase 4 multi-year program and are not claimed by the present synthesis.

With the v0.21 cosmetics phase shipped in May 2026, the protocol's empirical anchor base spans five substrate families: kitchenware (v0.16 kitchen knives, v0.17 premium kitchenware), indie fragrance (v0.18), audiophile electronics (v0.19), skincare (v0.20), and cosmetics (v0.21). Six pre-registered measurement events between March and May 2026. The five families span three distinct consumer-discovery environments: editorial-authority categories where canonical recommendation drives retrieval (skincare, kitchenware), community-curated categories where forum and review-channel discourse drives retrieval (audiophile, indie fragrance), and social-media-native categories where celebrity and viral discourse drives retrieval (cosmetics). The construct's verdicts hold across all three environments.

The reference panel is six LLMs held fixed across the program: Claude Opus 4.5, Claude Sonnet 4.5, GPT-4o, GPT-4o-mini, Gemini 2.5 Flash, and Gemini 2.5 Flash Lite. Holding the panel fixed across phases is the program's cross-phase comparability discipline — verdicts are properties of brand × substrate × panel triples measured under the same set of intermediaries each time. Per-LLM rankings and provider-level performance comparisons are not the report's subject; the construct is a property of the brand × substrate × panel system as a whole.

This report consolidates the five-family anchor base under five propositions. P1: AI Availability is now a measurable third channel of brand presence, anchored across five substrate families. P2: off-panel brand presence is real and its channel signature tracks Identity Load — Estée Lauder and Clinique surface canonical-pure, Glossier surfaces cultural-pure. P3: Recall splits into a canonical channel and a cultural channel that can be managed separately — Rare Beauty's 1:17 R_cat:R_cult split is the textbook diagnostic. P4: Identity Load predicts which channel will lead, at confidence-interval rigor. P5: methodology v1.6 closes the operational layer that makes P2 through P4 measurable.

The v0.21 cosmetics phase produced the program's strongest single-phase evidence base. Three primary findings stack: cultural-channel Recall populated the Type 2 dissociation quadrant above the EMERGED threshold for the first time (three Cell B cases — Rare Beauty, Huda Beauty, Kylie Cosmetics); the Identity-Load moderator returned CONFIRMED with the program's first CONFIRMED moderator verdict at any layer (95% bootstrap confidence intervals satisfying the monotonic-gradient check across cells); and Phantom Brand Persistence returned CONFIRMED with margin (six off-panel brands clearing the persistence threshold; the Glossier validity anchor passing at R_phantom = 12).

Three of the six phantom brands in cosmetics are channel-pure. Estée Lauder appears exclusively in canonical-channel frames (R_cat = 13, R_cult = 0); Clinique also exclusively canonical; Glossier exclusively in cultural-channel frames (R_cat = 0, R_cult = 12). The channel signature tracks Identity Load — prestige and heritage cosmetics in the authority pathway; celebrity, DTC, and cult brands in the discourse pathway. A brand's AI Availability cannot be inferred from absence on any single measurement panel; the phantom layer must be measured against an exogenously-constructed reference vocabulary at the substrate level.

Rare Beauty is the textbook diagnostic case. With Recognition saturated at C_P = 6/6 (universally categorized as a cosmetics brand by every panel intermediary), the brand's canonical-channel Recall is 1/18 while its cultural-channel Recall is 17/18 — a 1:17 split that establishes the upper bound on how decoupled the two channels can be within a single brand. The diagnostic is load-bearing for brand-strategy practice: a brand with this profile has identifiably different brand-strategy options than a brand with the inverse profile, and the protocol's measurement procedure surfaces the distinction at the audit layer rather than leaving it to interpretation.

For brand practice, the report's most actionable finding is the two-channel decomposition. R_cat (canonical Recall) and R_cult (cultural Recall) respond to disjoint inputs and can be managed separately. Canonical Recall responds to authority surfaces — editorial coverage, expert recommendation, category-best curation, professional certification. Cultural Recall responds to discourse density — social-media volume, celebrity endorsement, viral content, community-curated cult-tier discourse. Practitioners can audit a brand's current channel position with the protocol's measurement procedure and target the underperforming channel with the appropriate intervention class without conflating the two pathways into a single Recall budget.

The report's claims are scoped by design. AIAS™ 1.0 names the Presence component of the AIAS construct, not the full six-component composite. Construct validity (predictive validity against behavioral outcomes; convergent and discriminant validity against Mental and Physical Availability) is Phase 3 future work. The consumer-behavior correlate of AI Availability — whether AI Availability scores predict consideration, search, or purchase behavior — is plausible on theoretical grounds but has not been measured by the present program. The report establishes measurability under pre-registration discipline across five substrate families; validation against behavioral outcomes and integration of the remaining five composite components is the work the next decade of measurement-program research will do.

What we measured

AIAS™ measures AI Availability through a pre-registered protocol that runs the same two-phase measurement against each new substrate family under a panel of six LLMs held fixed across the program. The protocol's specifications are locked at git tag at each new phase before any acquisition; the methodology version that governs scoring is itself locked at git tag before retrospective scoring is applied to any prior-phase corpus. The methodology version that governs the present synthesis is v1.6 (SSRN 6816340, May 2026).

The reference panel is six LLMs locked at v0.17 and held constant across the program: Claude Opus 4.5, Claude Sonnet 4.5, GPT-4o, GPT-4o-mini, Gemini 2.5 Flash, and Gemini 2.5 Flash Lite. Holding the panel fixed across phases is the program's cross-phase comparability discipline — verdicts produced in any one phase are evaluable against verdicts from any other phase because the measurement instrument is the same. Per-LLM rankings and provider-level performance comparisons are not the program's subject; verdicts are properties of brand × substrate × panel triples evaluated under the same set of intermediaries each time.

Phase A — Recognition. Each brand in the panel registry is presented to each of the six panel intermediaries with a category-membership probe (template locked at v1.4 specification): the model is asked whether the brand is commonly recognized as a member of the substrate's category, with a yes/no response. The brand's Recognition score C_P is the count of yes responses across the six panel intermediaries, range 0–6. C_P is the Recognition component of AI Availability — a brand with C_P = 6 is universally categorized by the panel; a brand with C_P = 0 is not categorized as a category member by any panel intermediary.

Phase B — Recall, two-channel decomposition. Each panel intermediary receives a six-frame open-ended query battery against the substrate's category, and the panel's responses are scanned for brand mentions against the locked substrate registry. v1.5 introduced the two-channel decomposition: three frames anchor the canonical channel (R_cat — best brands, most-recommended brands, highest-quality brands; max 18 = 3 frames × 6 models) and three frames anchor the cultural channel (R_cult — most popular brands, celebrity-favorite brands, viral or cult-favorite brands; max 18). The two channels operate on disjoint inputs and can be managed separately at the brand-strategy intervention layer.

Three dissociation patterns at the v1.5 thresholds. Iwachu: high Recognition with sparse canonical Recall (C_P 5 R_cat 2) — named after the Japanese kitchenware brand that anchored the first case in v0.17. Type 1: canonical-channel-preferred (R_cat 5 R_cult 2) — prestige and authority-anchored brands cluster here. Type 2: cultural-channel-preferred (R_cat 2 R_cult 5) — celebrity-DTC and discourse-driven brands cluster here. The three patterns occupy the Recognition × Recall plane and bracket the dissociation construct's empirical geometry. v0.21 cosmetics is the first phase where Type 2 cleared the EMERGED threshold (three Cell B cases).

Methodology v1.6 — three increments. First, a substrate-level Recognition pre-screen identifies substrates where every panel brand is categorically recognized in every cell (the cosmetics case) and routes them to a distinct verdict state preserving the substrate's evidentiary value at the moderator and phantom layers. Second, an independent moderator pathway evaluates Identity-Load asymmetry as = mean(R_cult) − mean(R_cat) per cell with 95% bootstrap confidence intervals (n = 10,000) and a monotonic-gradient check across cells. Third, the Phantom Brand Persistence Phase B extension scores off-panel brand presence against an exogenously-constructed reference vocabulary (union of the panel, a pre-registered top-50 market-share list, and prior-phase Phase B emergents cross-validated by the market-share list) with persistence threshold K = 6 and a pre-registered validity anchor.

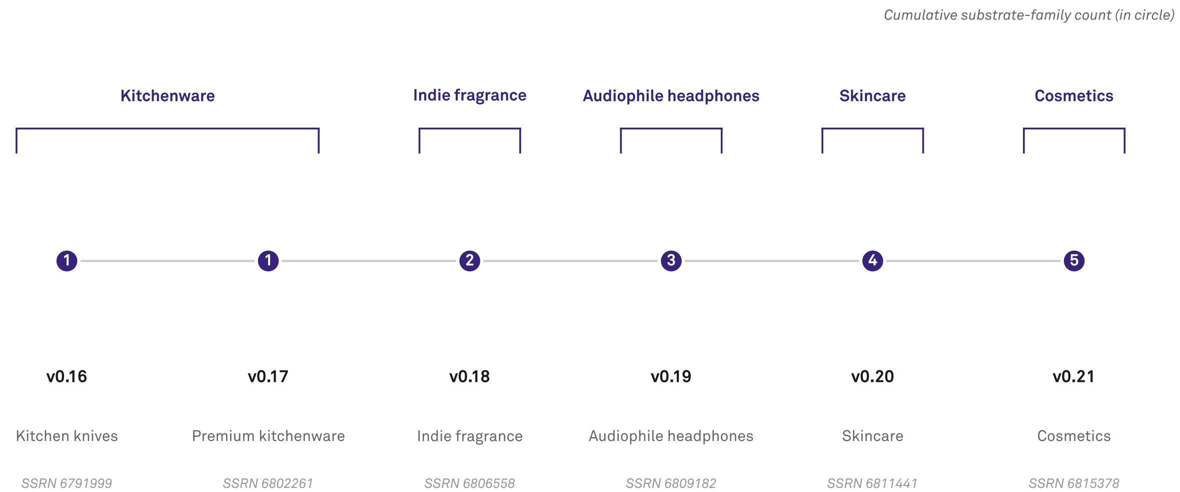
Five substrate families anchor the construct. v0.16 kitchen knives and v0.17 premium kitchenware (kitchenware family — Cell A heritage, Cell B Japanese-cult, Cell C mass); v0.18 indie fragrance (Cell A heritage / niche, Cell B perfumista-cult, Cell C mass); v0.19 audiophile electronics (two-cell uniform-IL design — Heritage and Boutique, with audiophile-community curation producing the substrate-shape case the IL-gradient framework cannot test against); v0.20 skincare (Cell A heritage / clinical, Cell B celebrity-DTC, Cell C mass / drugstore); v0.21 cosmetics (Cell A prestige, Cell B celebrity-DTC / cult, Cell C drugstore / mass). Each phase locked its 24-brand panel — eight brands per cell × three cells, or twelve per cell × two cells for v0.19 — at pre-registration before any acquisition.

Pre-registration discipline. Every phase enters the program under pre-registration. Panel construction, substrate registry, Phase A probe template, Phase B six-frame battery, hypothesis set, decision rules, and verdict matrices are locked ex-ante at a git tag (v0.NN-prereg-rN); the methodology version is locked at its own tag (v1.NN-prereg-rN). All artifacts — pre-registrations, acquisition data, scoring code, verdict outputs, reference vocabularies — are deposited under Open Science Framework project ec6wh (osf.io/ec6wh). The discipline is what allows the construct's verdicts to be falsifiable and the construct's measurability claim to stand or fall on its own evidence.

FINDING 01

The five-substrate empirical anchor base is complete. AI Availability is anchored, not provisional.

Cumulative substrate-family anchor base (v0.16 → v0.21)
Five substrate families anchored under locked v1.6 methodology



Source: AIAS™ 1.0 synthesis paper data — osf/aias_1_0/aias_1_0_synthesis_data.json (lock tag aias-1-0-data-locked). Upstream: SSRN 6791999 (v0.16), 6802261 (v0.17), 6806558 (v0.18), 6809182 (v0.19), 6811441 (v0.20), 6815378 (v0.21); methodology v1.6 SSRN 6816340. Five methodology papers v1.2–v1.6 in citation chain.

Figure 1 · Cumulative substrate-family anchor base, v0.16 → v0.21. Six pre-registered measurement events between March and May 2026 anchor the AIAS™ Presence Measurement Protocol across five substrate families: kitchenware (v0.16 + v0.17), indie fragrance (v0.18), audiophile electronics (v0.19), skincare (v0.20), and cosmetics (v0.21). All five families measured under one locked methodology version (v1.6, SSRN 6816340). The anchor-base completion changes the construct's standing from one-or-two-substrate-anchored to cross-category measurable under pre-registration discipline.

Six pre-registered measurement events between March and May 2026 anchor the AIAS™ Presence Measurement Protocol across five substrate families: kitchenware (v0.16 + v0.17), indie fragrance (v0.18), audiophile electronics (v0.19), skincare

(v0.20), and cosmetics (v0.21). Each phase locked its panel, brand registry, probe wording, hypothesis set, decision rules, and verdict matrices at a git tag before any measurement data was acquired. The methodology version itself is locked at v1.6 (SSRN 6816340, May 2026), with the locked-methodology boundary applied to retrospective scoring per the protocol's published rules. Holding both the panel and the methodology fixed across phases is the program's cross-phase comparability discipline — verdicts produced in any one phase are evaluable against verdicts from any other phase, because the measurement instrument is the same. The construct stands or falls on this discipline.

The five families were not selected for breadth alone; each contributed a methodology-relevant observation. Kitchenware (v0.16 + v0.17) established the original Recognition × Recall dissociation pattern — named Iwachu after the Japanese kitchenware brand that anchored the first case — and surfaced the substrate-language-carryforward phenomenon that constrains panel design for non-English-language substrates. Indie fragrance (v0.18) extended the dissociation framework into an English-language perfumistas-native discourse environment and produced the first multi-cell Iwachu generalization, establishing that the pattern is not specific to a single substrate family or discourse language. Audiophile electronics (v0.19) was the substrate-shape case — a uniform-Identity-Load substrate that the IL-gradient framework cannot test against, surfacing as the methodology-design counterexample that motivated the v1.6 substrate Recognition pre-screen. Skincare (v0.20) was the first prospective phase under the v1.5 two-channel Recall design and produced the first PARTIAL Type 2 verdict under the new framework, plus a skincare-specific Cell A architecture finding (clinical authority lives in a different place than prestige authority). Cosmetics (v0.21) is the program's strongest single-phase evidence base, producing the three v0.21 headline findings that propositions P2 through P4 carry below.

One observation surfaces under the synthesis view that no single phase's verdict matrix could surface within its own scope. The Type 2 dissociation pattern — cultural-channel-preferred Recall, defined as R_cat ≥ 2 and R_cult ≥ 5 — was pre-registered to be tested in each phase against the cell where Identity Load is highest (Cell B in the standard panel design). When the all-cell algorithmic dissociation framework is applied uniformly to the cross-phase data rather than to each phase's Cell-B-anchored verdict scope, two out-of-cell Type 2 cases recover. La Mer (v0.20 skincare, Cell A — prestige; C_P = 6, R_cat = 2, R_cult = 7) and e.l.f. Cosmetics (v0.21 cosmetics, Cell C — drugstore/mass; C_P = 6, R_cat = 1, R_cult = 9) jointly establish out-of-cell Type 2 as a recurrent feature of the dissociation pattern across substrate families. The per-phase published verdicts (v0.20 PARTIAL, v0.21 EMERGED) stand as authoritative within their pre-registered Cell-B scope; the cross-phase view adds a substrate-wide pattern observation that practitioners reading the synthesis can carry forward without amending either phase paper's record. For brand managers: a brand can occupy a Type 2 channel position regardless of the panel-design cell it was placed in — supply-side tier classification (prestige, mass) does not determine where in the two-channel space the brand will land.

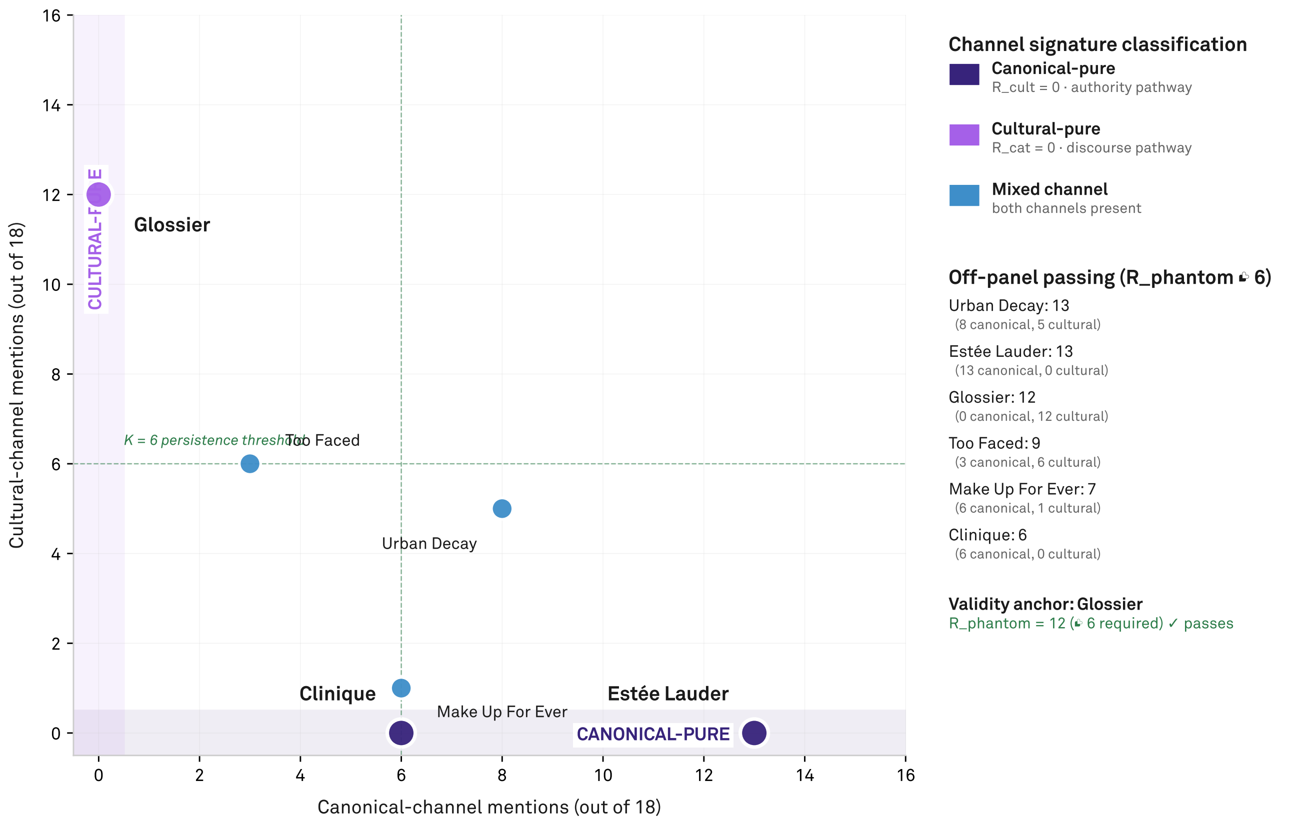
For brand managers, the anchor-base completion changes the construct's standing. AI Availability has graduated from a construct claimed on one or two substrate-family anchors to a construct that holds across five distinct families measured under one locked methodology. The construct can be applied to new substrate families as candidates for prospective measurement — v0.22 and successor phases under the same v1.6 lock — and brand-strategy practitioners can consume the construct as a stable measurement instrument rather than a moving-target methodology. Two boundaries persist. Construct validity — whether AI Availability scores predict downstream consumer behavior such as consideration, search, or purchase — is Phase 3 future work and is not established by the present anchor base. The full six-component AIAS™ composite (Ranking, Consistency, Coverage, Grounding, Sentiment) is reserved for a Phase 4 multi-year program. The present claim is bounded to Presence-component measurability across five substrate families, and the program's credibility rests on that bounded claim rather than on an over-claimed composite the data does not yet carry.

FINDING 02

Brand presence in AI retrieval extends beyond the panel. Channel signature tracks Identity Load.

Off-panel cosmetics brands — channel signature

Three of six phantom brands channel-pure. Estée Lauder and Clinique canonical-pure; Glossier cultural-pure.



Source: v0.21 cosmetics measurement — osf/aias_1_0/aias_1_0_synthesis_data.json (lock tag aias-1-0-data-locked). v1.6 Phantom Brand Persistence specification, SSRN 6816340.

Figure 2 · Off-panel channel signature on v0.21 cosmetics. Six off-panel brands cleared the K = 6 persistence threshold; three of six are channel-pure. Estée Lauder (R_cat = 13) and Clinique (R_cat = 6) appear exclusively in canonical-channel frames — the authority pathway: prestige and heritage cosmetics. Glossier (R_cult = 12) appears exclusively in cultural-channel frames — the discourse pathway: celebrity, DTC, and cult brands. The channel signature tracks the brands' Identity Load, with the same gradient operating on off-panel brands as on the panel-internal brands (Finding 04). Glossier was pre-registered as the validity anchor and passed at R_phantom = 12, well clear of the K = 6 threshold.

On the v0.21 cosmetics measurement, six off-panel brands surfaced persistently in the panel's category-anchored responses. Each cleared the K = 6 persistence threshold under the v1.6 Phantom Brand Persistence specification: Urban Decay (R_phantom = 13), Estée Lauder (13), Glossier (12), Too Faced (9), Make Up For Ever (7), and Clinique (6). The brands were absent from the v0.21 panel — the panel was tier-balanced at 8 brands per cell × 3 cells = 24 brands, drawn from prestige, celebrity-DTC/cult, and drugstore/mass tiers — but appeared in the panel intermediaries' responses to category-anchored prompts at sufficient frequency to meet the persistence threshold. The phantom-layer construct is the program's first measured component for off-panel presence: prior to v1.6, off-panel mentions were noted as a descriptive observation across multiple phases but not scored as a measurement component. v0.21 is the substrate where the construct was empirically motivated and where the calibration was anchored.

Three of the six phantom brands are channel-pure. Estée Lauder appears exclusively in canonical-channel frames (R_cat = 13, R_cult = 0); Clinique also exclusively canonical (R_cat = 6, R_cult = 0); Glossier appears exclusively in cultural-channel frames (R_cat = 0, R_cult = 12). The other three carry mixed signatures consistent with their brand positioning: Urban Decay leans canonical (R_cat = 10, R_cult = 3); Too Faced is split (R_cat = 5, R_cult = 4); Make Up For Ever leans canonical (R_cat = 6, R_cult = 1). The channel-pure cases sort the brands by Identity Load — prestige and heritage cosmetics (Estée Lauder, Clinique) in the authority pathway; cult and DTC cosmetics (Glossier) in the discourse pathway. The same Identity-Load gradient that operates on the panel-internal brands (Finding 04 below) operates on the off-panel brands. The phantom layer is not panel-incidental; it is substrate-structural.

For brand-audit practice, the phantom layer is the structural reason a brand's AI Availability cannot be inferred from absence on any single measurement panel. A brand not on the panel can nonetheless drive AI retrieval at measurable frequency, and its channel signature carries information about how the brand is being retrieved by which class of prompt frame. Audit procedures that rely on panel-internal scoring alone systematically miss the phantom layer; the construct must be measured against an exogenously-constructed reference vocabulary at the substrate level. The reference vocabulary for v0.21 cosmetics was constructed as the union of the panel, a pre-registered top-50 market-share list, and prior-phase Phase B emergents cross-validated by the market-share list — designed to break the endogeneity hazard that discovery-driven vocabulary expansion would introduce. Glossier was pre-registered as the validity anchor (its strong off-panel presence in v0.20 skincare and earlier-phase cosmetics observation made it the program's a priori expected highest R_phantom) and passed at R_phantom = 12, well clear of the K = 6 threshold. The construct meets its pre-registered validity check on the calibration substrate.

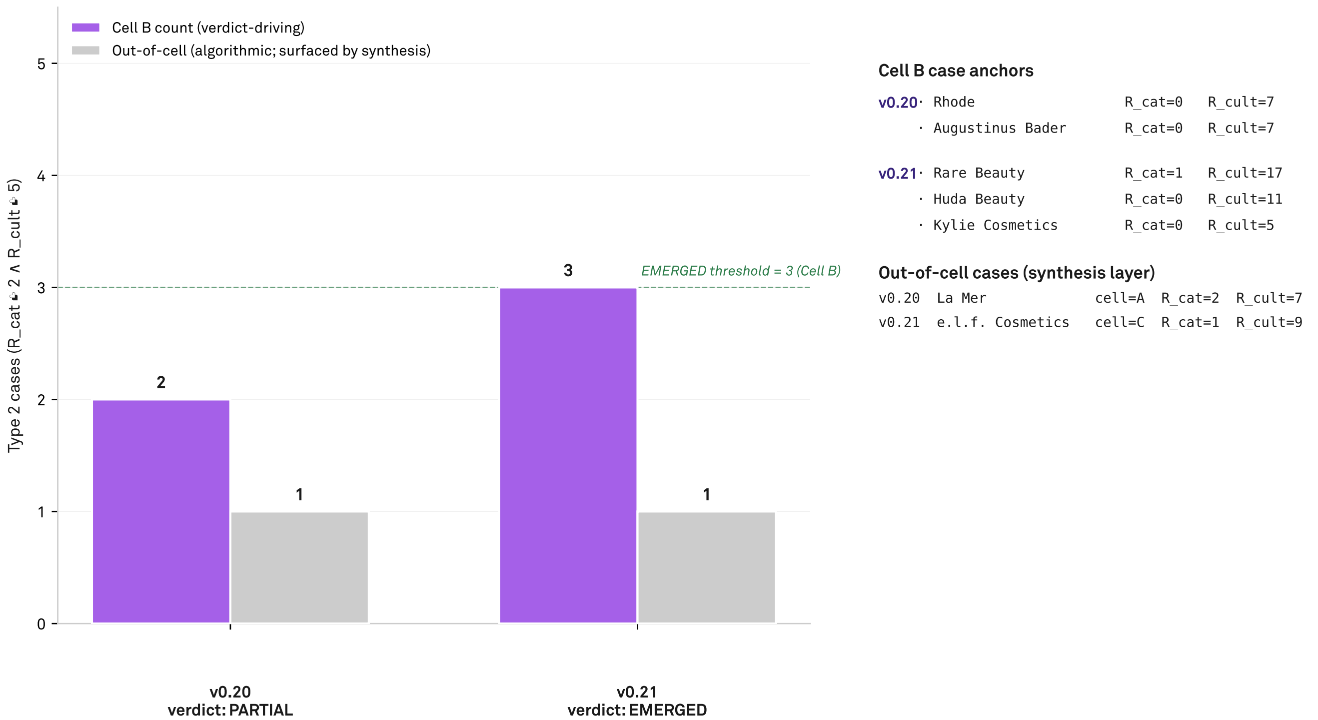
The Phantom Brand Persistence construct is calibrated on cosmetics, not yet replicated across substrates. v0.21 is the substrate where the construct was empirically motivated, where the reference vocabulary was constructed, and where the validity anchor passed. Cross-substrate replication is the genuine generalization test, and that test is v0.22+ future work — prospective phases under the locked v1.6 protocol measuring phantom-layer behavior on substrate families other than cosmetics. Brand managers reading this report should treat the phantom-layer finding as established for cosmetics, hypothesized for other substrate families, and pending prospective measurement in each new substrate where the construct's behavior matters for brand-strategy decisions. The construct's value as a managerial instrument is highest in substrates where the brand-of-interest's panel-membership status is itself a moving question: emerging brands, indie tiers, and substrate categories where panel construction is contested. Practitioners can identify high-phantom-risk categories by checking whether category-best lists vary substantially across editorial sources or whether community-curated emergents regularly enter category discourse without appearing on supply-side market-share leaderboards. For categories with stable panel construction and broad market consensus on category membership, the phantom layer matters less.

FINDING 03

Recall splits into two channels that can be managed separately. Rare Beauty 1:17 is the textbook diagnostic.

Type 2 quadrant emergence — v0.20 PARTIAL → v0.21 EMERGED

Cell B count drives the H_Type2_emergence verdict (≥3 → EMERGED); cross-phase all-cell count surfaces out-of-cell cases



Source: AIAS™ 1.0 synthesis paper data — osf/aias_1_0/aias_1_0_synthesis_data.json (lock tag aias-1-0-data-locked). Upstream: SSRN 6791999 (v0.16), 6802261 (v0.17), 6806558 (v0.18), 6809182 (v0.19), 6811441 (v0.20), 6815378 (v0.21); methodology v1.6 SSRN 6816340. Five methodology papers v1.2–v1.6 in citation chain.

Figure 3 · Type 2 quadrant cross-phase emergence: v0.20 PARTIAL → v0.21 EMERGED. The Type 2 cultural-channel-preferred quadrant (R_cat ≤ 2 ∧ R_cult ≥ 5) cleared the EMERGED threshold for the first time at v0.21 cosmetics with three Cell B cases: Rare Beauty (R_cat = 1, R_cult = 17 — the program's textbook Recognition × Recall dissociation case), Huda Beauty (0:11), and Kylie Cosmetics (0:5). v0.20 skincare returned PARTIAL on the same test with two Cell B cases below the EMERGED threshold of 3. Two cross-phase out-of-cell Type 2 anchors — La Mer (v0.20 Cell A) and e.l.f. Cosmetics (v0.21 Cell C) — establish out-of-cell Type 2 as a recurrent feature of the dissociation pattern across substrate families.

The Recall layer of the protocol splits into two channels at the v1.5 specification. The canonical channel (R_cat) is anchored by three prompt frames: best brands in the category, most-recommended brands in the category, highest-quality brands in the category. The cultural channel (R_cult) is anchored by three prompt frames: most popular brands in the category, celebrity-favorite or influencer-driven brands in the category, viral or cult-favorite brands in the category. Each frame is sent to each of the six panel intermediaries; per-brand R_cat and R_cult are mention counts ranging 0–18 (3 frames × 6 models). The two channels operate on disjoint inputs: R_cat responds to authority surfaces — editorial coverage, expert recommendation, category-best curation, professional certification, dermatologist endorsement, makeup-artist recommendation. R_cult responds to discourse density — social-media volume, celebrity endorsement, viral content, community-curated cult-tier discourse, founder-identity association. Conflating them under a single Recall metric loses the actionable diagnosis: a brand strong on one channel and weak on the other reads as average under single-channel scoring but reads as channel-asymmetric under the two-channel decomposition, and the asymmetry is the audit signal.

Rare Beauty is the textbook diagnostic case in v0.21 cosmetics. With Recognition saturated at C_P = 6/6 — universally categorized as a cosmetics brand by every panel intermediary — the brand's canonical-channel Recall is 1/18 while its cultural-channel Recall is 17/18. The 1:17 split establishes the upper bound on how decoupled the two channels can be within a single brand: nearly every cultural-frame mention available, almost none of the canonical-frame mentions. Two Cell B companions joined Rare Beauty in clearing the EMERGED threshold for the Type 2 cultural-channel-preferred quadrant (R_cat ≤ 2 and R_cult ≥ 5): Huda Beauty (R_cat = 0, R_cult = 11) and Kylie Cosmetics (R_cat = 0, R_cult = 5). The three Cell B cases share a brand-positioning profile — celebrity-DTC cosmetics with founder-identity association at the substrate's high-Identity-Load tier — and the Type 2 emergence is the empirical signature of that positioning class translating to AI retrieval.

For brand strategy, the two-channel decomposition makes interventions targetable rather than budget-consuming. A brand whose AI Availability audit returns low canonical-channel Recall has identifiable intervention pathways: build editorial-coverage relationships in the category's authority press; pursue makeup-artist and dermatologist recommendations where the category supports them; secure category-best list inclusion at curated review surfaces; target professional-certification visibility. A brand whose audit returns low cultural-channel Recall has a different intervention class: invest in social-media volume; secure celebrity or influencer endorsement; build community-curated cult-tier discourse through founder identity and brand storytelling. A brand strong on one channel and weak on the other can target the weaker channel without trading off against the stronger; the two channels do not share input budgets and they do not compete for the same brand-strategy resources. The audit is the diagnostic, not the prescription — but the audit surfaces a structural distinction (canonical vs. cultural channel position) that the prescription can then act on without the brand-strategy team having to negotiate the intervention class as a separate question.

Rare Beauty's 1:17 channel split is the upper bound; the dissociation framework also identifies the inverse pole. The Type 1 pattern — canonical-channel-preferred Recall, R_cat ≥ 5 and R_cult ≤ 2 — surfaces in the v0.21 cosmetics Cell A (prestige) with Bobbi Brown (R_cat = 8, R_cult = 0) and Laura Mercier (R_cat = 5, R_cult = 0) as the channel-pure canonical anchors. The Type 1 and Type 2 poles bracket the two-channel space within a single substrate. Brand managers auditing a candidate brand can locate its position in the R_cat × R_cult plane and read its position against the substrate's Type 1 and Type 2 anchors: a brand near a pole has identifiable intervention options matched to the dominant channel's input class; a brand near the Iwachu region — high Recognition with low Recall in both channels — has a different question — why the categorical Recognition isn't converting to either channel of Recall; a brand in the middle reads as positioned-but-undifferentiated and may warrant a positioning decision before an AI Availability intervention. The two-channel decomposition turns AI Availability from a single metric into an audit surface with structural geometry.

FINDING 04

Identity Load predicts the AI channel where a brand will surface. Cosmetics returned the program's first CONFIRMED moderator verdict.

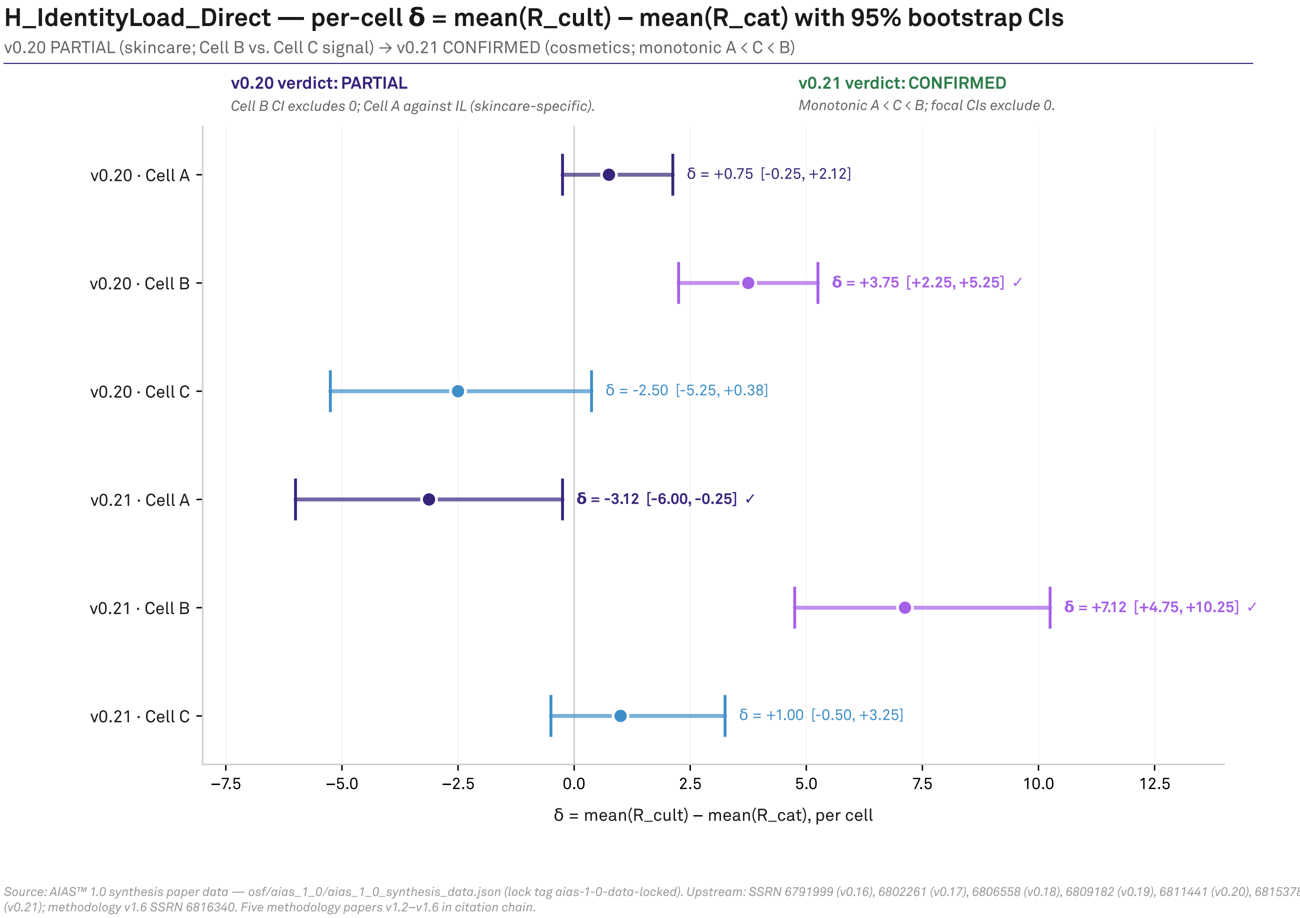


Figure 4 · Identity-Load moderator □ values across v0.20 + v0.21 with 95% bootstrap confidence intervals. □ = mean(R_cult) – mean(R_cat) per cell; positive values indicate cultural-channel lead, negative values indicate canonical-channel lead. v0.21 cosmetics returned the program's first CONFIRMED moderator verdict at any layer: Cell B □ = +7.13 (CI [+4.75, +10.25]) excludes zero in IL-predicted direction; Cell A □ = −3.13 (CI [−6.00, −0.25]) excludes zero in opposite direction as IL gradient predicts; Cell C □ = +1.00 (CI [−0.50, +3.25]) between Cell A and Cell B, satisfying monotonic-gradient check. v0.20 skincare PARTIAL on the same test with substrate-specific Cell A architecture (clinical authority lives in mass tier, not prestige).

Identity Load is the substrate-design property that distinguishes the panel's cells: prestige (Cell A, medium IL — established editorial recognition and category-best curation as primary brand identity); celebrity-DTC / cult (Cell B, high IL — strong founder-identity association and community-curated discourse as primary brand identity); drugstore / mass (Cell C, low IL — distribution-led brand identity, less identity-anchored discourse). The v1.6 Increment 2 specification (SSRN 6816340) introduced the IL Direct test: per cell, = mean(R_cult) – mean(R_cat) is computed with a 95% percentile bootstrap confidence interval (n = 10,000), and the verdict matrix requires the CI to exclude zero in the IL-predicted direction in Cell B (the high-IL anchor, cultural-channel-leading expected); the IL-predicted opposite direction in Cell A (the medium-IL anchor, canonical-channel-leading expected); and the monotonic-gradient check Cell A < Cell C < Cell B across cells. The test is evaluable independently of the protocol's Regime 4 conditions — a separation that the v1.5 framework did not permit and that v1.6 introduced specifically to surface the moderator signal where prior framework versions could not.

The v0.21 cosmetics measurement returned CONFIRMED on the IL Direct test, with each cell's falling in its predicted region and the monotonic-gradient check satisfied. Cell B = +7.13 with 95% bootstrap CI [+4.75, +10.25] — the program's strongest cultural-channel-lead signal, with cultural-frame mentions outpacing canonical-frame mentions by 7.13 average across Cell B brands. (Rare Beauty's 1:17 split from Finding 03 is the Cell B exemplar.) Cell A = −3.13 with CI [−6.00, −0.25] — the canonical channel leads in the medium-IL prestige cell, as the IL gradient predicts. Cell C = +1.00 with CI [−0.50, +3.25] — between Cell A and Cell B, satisfying the monotonic-gradient check. v0.21 is the program's first CONFIRMED moderator verdict at any layer. The IL-gradient construct moves from a hypothesized pattern visible in earlier-phase single-channel data to a measured moderator with confidence-interval rigor.

The skincare measurement (v0.20) returned PARTIAL on the same test, with the substrate's IL signal carrying directionally with prediction but with substrate-specific architecture. Cell B = +3.75 with CI [+2.25, +5.25] cleared the IL-predicted direction. Cell A = +0.75 with CI [−0.25, +2.12] carried against prediction with CI overlapping zero — clinical and heritage skincare brands such as CeraVe, La Roche-Posay, and Eucerin do not concentrate in canonical-channel Recall the way prestige cosmetics brands do, because skincare's canonical authority lives in dermatologist-recommended and clinical-result-anchored discourse rather than in editorial-prestige curation. Cell C = −2.50 with CI [−5.25, +0.38] broke the monotonic-gradient check with Cell A. The skincare finding is substrate-specific rather than methodology-disqualifying: the IL-gradient model holds for the substrate, but the cell where canonical authority concentrates is not the cell that the standard panel-design heuristic predicts. Indie fragrance (v0.18) carried an IL-gradient signature visible in the v1.4 single-channel data and articulated the moderator hypothesis as a coherent cross-phase question — the v1.5 two-channel decomposition was designed in response, and the v1.6 IL Direct test was specified to surface the signal more cleanly than the v1.5 framework could.

For brand strategy, the IL-gradient moderator gives the audit a predictive prior. A brand's cell position (and the IL construction of that cell) carries an expected channel asymmetry: high-IL cells expect cultural-channel lead; medium-IL cells expect canonical-channel lead; low-IL cells lie between under monotonic-gradient discipline. A brand whose measured channel asymmetry matches the cell's expected asymmetry reads as cell-typical; deviation from the expected asymmetry is itself diagnostic information — the brand is operating against the cell's IL grain, and the diagnostic surfaces the deviation as a question worth investigating (a high-IL cell brand with unexpectedly strong canonical Recall may be punching above its cell's editorial weight; a low-IL cell brand with unexpectedly strong cultural Recall may be operating with social-media leverage uncharacteristic of its tier). One boundary is essential: Identity Load is operationalized through the substrate's panel design, not through independently-measured consumer perceptions of the brand. The moderator operates on the panel-design construction, not on a separately-measured consumer-side construct of brand identity. The construct's mechanism is the IL-gradient property of the substrate-as-measured, and the brand-strategy implication is bounded to substrates where the panel-design IL classification meaningfully tracks the brand-of-interest's category positioning.

FINDING 05

Methodology v1.6 closes the operational layer. Three increments shipped May 2026.

Protocol v1.6 (SSRN 6816340, May 2026) shipped three increments that closed three measurement gaps the prior framework versions could not address. The first increment is a substrate-level Recognition pre-screen: a substrate whose Phase A Recognition distribution is uniformly saturated — every brand recognized as a category member by every panel intermediary, in every cell — is routed to a distinct verdict state rather than collapsed into the prior framework's FALSIFIED bucket. The second increment is an independent moderator pathway: the Identity-Load test is evaluable per-cell with bootstrap confidence intervals, separately from the protocol's four-regime test that the prior framework coupled it to. The third increment is the Phantom Brand Persistence Phase B extension: off-panel brand presence is scored against an exogenously-constructed reference vocabulary with a persistence threshold and a validity anchor, lifting the construct from a descriptive side observation to a measured component. Each increment was specified in v1.6's methodology paper and pre-registered at git tag v1.6-prereg-r1 before retrospective scoring was applied to the v0.16–v0.21 corpus.

Each increment found its first measured anchor in v0.21 cosmetics. Increment 1 returned its first uniform-saturation trigger: every panel brand in every cell scored C_P = 6/6, producing distinct C_P count = 1 and modal share = 1.000 in Cell A, Cell B, and Cell C — a substrate where the prior framework's four-regime test was not applicable, and where v1.6's REGIME-4-UNAVAILABLE-AT-RECOGNITION verdict preserved the substrate's evidentiary value at the moderator and phantom layers (the substrate would have read as a FALSIFIED bucket under prior framework versions, losing the discrimination signal v1.6's pre-screen surfaces). Increment 2 returned the program's first CONFIRMED moderator verdict at any layer, with Cell B = +7.13 (CI [+4.75, +10.25]), Cell A = −3.13 (CI [−6.00, −0.25]), Cell C = +1.00 (CI [−0.50, +3.25]), monotonic-gradient check satisfied. Increment 3 returned CONFIRMED with margin: six off-panel brands cleared the K = 6 persistence threshold, and the pre-registered Glossier validity anchor passed at R_phantom = 12. Together they form a coordinated audit pathway for substrates where prior framework versions would have collapsed the verdict — Increment 1 preserves the substrate's evidentiary value, and Increments 2 and 3 carry the discrimination signal that Recognition can't supply when it's exhausted at ceiling.

For the brand-strategy practitioner, v1.6's increments translate to vocabulary the prior framework versions could not provide. A substrate where every brand is categorically recognized is now identifiable as a category where the discrimination signal lives entirely in Recall — a vocabulary distinction that lets brand managers diagnose maximally-coded categories (cosmetics is the first measured example; other consumer-facing categories with broad media coverage and stable category-membership consensus are candidates for similar shape) without conflating them with categories where Recognition itself carries the diagnostic signal. The moderator pathway makes Identity-Load asymmetry an audit input, not an inferred property — the audit returns values with bootstrap CIs, and brand managers can compare a brand's cell-relative position against the cell's measured asymmetry directly. The phantom layer measures off-panel presence, which means brand audits for emerging brands not yet on standard category-tracking panels can include AI Availability as a measurable construct rather than treating panel-absence as missing data. v0.22 and successor phases apply v1.6 prospectively to new substrate families under the same locked protocol; the operational layer is what each new prospective phase consumes, not what each new phase has to first redefine.

AIAS™ 1.0 names the Presence component of the AIAS construct, not the full multi-component composite. The Tri-System framework names the composite as a 0–100 score across six measurable dimensions — Presence, Ranking, Consistency, Coverage, Grounding, Sentiment — and positions the composite as a multi-year research arc. AIAS™ 1.0 operationalizes the first of those six dimensions; the remaining five ship under a Phase 4 successor program with specifications, verdict matrices, and pre-registration discipline reserved for that program's pre-registration round. The version-number arc is load-bearing: 1.0 names the moment when the construct's first component reaches falsifiable measurement against a cross-substrate anchor base, and the version increments that follow will track the measurement surface rather than the architectural ambition. The construct does not jump to 2.0 by claiming additional components without measuring them. For brand managers consuming the report's findings, the boundary is the program's credibility asset — what AIAS 1.0 measures is what it claims to measure, and the multi-year roadmap is signposted, not over-claimed.

Limitations

The reference panel is held fixed at the v0.17 six-slot specification across all phases of the synthesis: Claude Opus 4.5, Claude Sonnet 4.5, GPT-4o, GPT-4o-mini, Gemini 2.5 Flash, and Gemini 2.5 Flash Lite. Holding the panel fixed across phases is the program's cross-phase comparability discipline. It is also a measurement-window constraint: the v0.21 verdicts and all earlier-phase verdicts are properties of brand × substrate × panel triples evaluated at acquisition time, not permanent properties of the substrate. Provider model substitutions over future-phase horizons are expected, and the construct's stability across model generations is itself future-work to demonstrate. Brand managers reading verdicts should treat them as measurements anchored in a specific panel state, with the panel's evolution itself a subject for the program's continued discipline.

Four of the five substrate families operated in English-language consumer-discovery environments. The kitchenware family's substrate-language carryforward at v0.16 — where Japanese-language category-bound discourse surfaced different brand mention patterns than English-language responses for the same substrate — is the program's only cross-language acquisition. Cross-language replication is future work. The construct may behave structurally differently in substrates where consumer discovery operates in non-English discourse environments, and the panel intermediaries' English-language training bias is a known structural feature of the measurement system. Brand managers in markets where the brand-of-interest's consumer-discovery surface is non-English should treat the report's findings as suggestive rather than directly transferable, pending prospective cross-language replication phases.

Cell-classification limits surfaced through the e.l.f. Cosmetics out-of-cell Type 2 case noted in Finding 01 ¶3 and Finding 03 ¶4. The IL-gradient cell structure assumes brands behave per their supply-side tier classification — a drugstore/mass brand should sit in Cell C, a celebrity-DTC brand in Cell B, a prestige brand in Cell A — and a brand whose observable channel behavior crosses the supply-side / discourse-side boundary exposes friction in the classification scheme. e.l.f. Cosmetics has drugstore/mass price points (Cell C by panel design) but a social-media-native cultural footprint that produces Type 2 dissociation more characteristic of Cell B brands. The classification scheme remains useful at the aggregate level — the IL gradient holds for cell-typical brands — but brand managers auditing brands at the supply-side / discourse-side boundary should expect classification friction.

The synthesis honors the v1.6-locked retrospective-scoring boundary without extension. Increment 1 (substrate Recognition pre-screen) classifies all six phases by narrative or by scoring, depending on the phase's data shape. Increment 2 (IL Direct) is bounded to v0.20 and v0.21 — the only phases that acquired R_cult data under the v1.5 two-channel design. Increment 3 (Phantom Brand Persistence) is bounded to v0.21, the substrate where the construct was empirically motivated and where the validity anchor was anchored. The boundary is methodology discipline, not data — extending Increments 2 and 3 retrospectively beyond their stated scope would require either retrospective channel re-mapping (which the program has explicitly disallowed) or post-hoc reference-vocabulary construction (which would compromise the validity anchor's pre-registration). Brand managers should read v0.16–v0.19 verdicts as v1.4-era authoritative within-phase results, with v1.6 framing added only where the retrospective scope supports it.

The synthesis claims measurability of the Presence component across five substrate families. It does not claim construct validity. Predictive validity against behavioral outcomes (whether AI Availability scores predict consideration, search, or purchase behavior), convergent validity across alternative panel constructions (whether other panel compositions yield the same brand-rank ordering), and discriminant validity against Mental and Physical Availability (whether AI Availability captures variance the existing constructs do not) are Phase 3 future work and are not established here. The consumer-behavior correlate of AI Availability is plausible on theoretical grounds — the Ehrenberg-Bass canon's logic implies that availability constructs should be load-bearing for category buying — but the present program has not measured the correlation. Brand managers should treat AI Availability as a measurement-program construct with anchored measurability and bounded scope, not as a predictor of downstream consumer behavior pending Phase 3 evidence.

What's next

The immediate next deliverable is v0.22, the program's first prospective phase under the locked v1.6 protocol. Substrate selection is open: candidates include automotive (a substrate with a different IL-gradient shape — brand identity carries strong individual-purchase identity but the discovery surface is dominated by review-channel discourse), consumer electronics outside audio (where heritage, cult, and mass tiers may map cleanly to the standard panel design), or premium spirits (where editorial authority and cultural-cult dimensions both run strong). Each candidate would test v1.6's three increments prospectively against a substrate not in the present retrospective scope. Pre-registration locks the panel and verdict matrices before acquisition, as the program's standing discipline requires.

Phase 3 is construct validation. Predictive validity against behavioral outcomes — whether AI Availability scores predict consumer behavior such as consideration, search, or purchase — is the discipline's primary validation criterion and the program's Phase 3 anchor. The Phase 3 design requires a behavioral-outcome dataset paired with AI Availability scores measured at the same brand × substrate boundary, against panel constructions held fixed for measurement comparability and varied for convergent-validity testing. Discriminant validity against Mental Availability and Physical Availability requires paired measurements of all three constructs against a common brand × substrate cohort. Both validation arcs sit in the Phase 3 pre-registration roadmap; neither is in the present synthesis's claim scope.

Phase 4 is the full six-component AIAS™ composite. The Tri-System framework names the composite as a 0–100 score across Presence (operationalized by the present synthesis), Ranking (which brand surfaces first), Consistency (how stable the ranking is across panel intermediaries), Coverage (how completely the panel surfaces the category's relevant brand set), Grounding (whether brand mentions are accompanied by category-relevant context), and Sentiment (how favorable the mention's affective framing is). The remaining five components ship under a Phase 4 successor program at the level of intent only; component-level prioritization, panel-design implications, and methodology successors will follow the same pre-registration discipline that v1.2 through v1.6 established for the Presence layer. The composite reaches a fully-measured 6.0 state only when all six components have been operationalized to the standard the Presence component reaches in the present synthesis.

Cross-language replication is the construct's second expansion axis. Non-English-language substrate phases — Japanese, Spanish, Mandarin, and others selected on substrate-availability grounds — will test whether the construct's measurement procedure transfers across discourse-language boundaries with the panel-internal scoring discipline held fixed. Panel-construction sensitivity is the third axis: studies that hold substrate and methodology version fixed while varying the panel — substituting providers, varying panel size, testing single-provider panels against multi-provider panels — will supply the convergent-validity evidence base that Phase 3 construct validation requires. The three expansion axes (substrate, language, panel-construction) together carry the construct's empirical anchor base from the present five families into a sustained replication program under the program's standing pre-registration discipline.

For brand managers planning against the construct's evolution, three timelines matter. AIAS 1.0's Presence component is anchored on cosmetics and four other families as of May 2026; new substrate families ship at the program's pace under v1.6 lock. Phase 3 construct validation is multi-year work that establishes whether AI Availability scores predict downstream consumer behavior — until that work ships, brand managers should treat AI Availability as a measurable property of brand × substrate × panel triples without inferring behavioral consequences. Phase 4 composite expansion is the longest-horizon arc; the version-number arc 1.0 → 6.0 will track the measurement surface rather than architectural ambition. Brand managers consuming the report should plan against the present anchored claim and against the future-work claims as separate planning tracks, with the construct's evolution timeline as the resource-allocation input.

Five propositions, anchored

Five propositions consolidate the five-substrate empirical anchor base into actionable framing for brand strategy. Each is anchored against the data the program has shipped under locked methodology v1.6. The table below states each proposition, the substantive evidence that anchors it, and the status as of the AIAS™ 1.0 milestone. The detailed interpretation of each — what it means, what it does not mean — follows in the section below the table.

P	Proposition	What the data shows	Status
P1	AI Availability is a measurable third channel of brand presence, anchored across five substrate families.	Five families measured under one locked methodology version (v1.6, SSRN 6816340): kitchenware, indie fragrance, audiophile electronics, skincare, cosmetics. Six pre-registered measurement events March–May 2026.	**ANCHORED**
P2	Off-panel brand presence is real and tracks Identity Load. Brands not on a measurement panel can still drive AI retrieval with auditable channel signatures.	Six off-panel cosmetics brands cleared K = 6. Estée Lauder + Clinique canonical-pure (R_cult = 0); Glossier cultural-pure (R_cat = 0). Glossier validity anchor passed at R_phantom = 12.	**CALIBRATED**
P3	Recall splits into a canonical channel (R_cat) and a cultural channel (R_cult) that operate on disjoint inputs and can be managed separately.	Rare Beauty textbook anchor: C_P = 6/6, R_cat = 1, R_cult = 17. Type 2 EMERGED with three Cell B cosmetics cases (Rare Beauty, Huda Beauty 0:11, Kylie Cosmetics 0:5).	**CONFIRMED**
P4	Identity Load predicts the AI channel where a brand will surface. Cell position carries an expected channel asymmetry that the audit can compare brand performance against.	v0.21 cosmetics returned the program's first CONFIRMED moderator at any layer. Cell B's cultural channel led by 7.13 mentions on average — the strongest IL-predicted asymmetry signal in the program. Cell A canonical-led by 3.13; Cell C balanced. All three cells matched the IL-gradient prediction with confidence-interval margin.	**CONFIRMED**
P5	Methodology v1.6 closes the operational layer. Three increments shipped May 2026 make P2 through P4 measurable.	Substrate Recognition pre-screen (Inc 1) + independent moderator pathway (Inc 2) + Phantom Brand Persistence (Inc 3). Each load-bearing for v0.21 verdicts.	**SHIPPED v1.6**

Proposition interpretation

Each proposition carries substantive interpretation beyond the table-row summary. The verdicts are what the data produced; the meaning of each verdict for the construct, the brand-strategy practitioner, and the v0.22+ trajectory is what the prose below addresses, structured as "What it means" + "What it doesn't mean" per proposition.

What it means. AI Availability has graduated from a construct claimed on one or two substrate-family anchors to a construct that holds across five distinct families measured under one locked methodology. The empirical base is sufficient for the program to apply the construct to new substrate families as candidates for prospective measurement, and for brand-strategy practitioners to consume the construct as a stable measurement instrument rather than a moving-target methodology. The five families span three distinct consumer-discovery environments (editorial-authority, community-curated, social-media-native), and the construct's verdicts hold across all three. **What it doesn't mean.** ANCHORED is not the same as construct validity. The synthesis claims measurability — that AI Availability can be measured under a falsifiable pre-registered protocol with cross-substrate replicability — not predictive validity against behavioral outcomes. Whether AI Availability scores predict consumer behavior (consideration, search, purchase) is Phase 3 future work and is not established by the present anchor base. The full six-component AIAS™ composite (Ranking, Consistency, Coverage, Grounding, Sentiment) is reserved for a Phase 4 multi-year program. The present claim is bounded to Presence-component measurability across five substrate families, and the program's credibility rests on that bounded claim rather than on an over-claimed composite the data does not yet carry.

What it means. A brand's AI Availability cannot be inferred from absence on any single measurement panel. The phantom layer surfaces off-panel brands at measurable frequency, with channel signatures that track the same Identity-Load gradient as the panel-internal brands. The construct's value as a managerial instrument is highest in substrates where the brand-of-interest's panel-membership status is itself a moving question — emerging brands, indie tiers, and substrate categories where panel construction is contested. Practitioners can identify high-phantom-risk categories by checking whether category-best lists vary substantially across editorial sources or whether community-curated emergents regularly enter category discourse without appearing on supply-side market-share leaderboards. **What it doesn't mean.** CALIBRATED is not the same as cross-substrate replicated. v0.21 cosmetics is the calibration anchor — the substrate where the construct was empirically motivated, where the reference vocabulary was constructed, and where the validity anchor (Glossier R_phantom = 12) passed. Cross-substrate replication is the genuine generalization test, and that test is v0.22+ future work — prospective phases under the locked v1.6 protocol measuring phantom-layer behavior on substrate families other than cosmetics. Brand managers reading this report should treat the phantom-layer finding as established for cosmetics, hypothesized for other substrate families, and pending prospective measurement in each new substrate where the construct's behavior matters for brand-strategy decisions.

What it means. R_cat and R_cult operate on disjoint inputs and can be managed separately. A brand with low canonical-channel Recall has identifiable intervention pathways in authority surfaces (editorial coverage, expert recommendation, category-best curation, professional certification); a brand with low cultural-channel Recall has a different intervention class in discourse density (social-media volume, celebrity endorsement, viral content, community-curated cult-tier discourse). Rare Beauty's 1:17 channel split establishes the upper bound on how decoupled the two channels can be within a single brand, and the dissociation framework identifies the inverse pole at the Type 1 anchors (Bobbi Brown 8:0, Laura Mercier 5:0 in v0.21 Cell A). **What it doesn't mean.** Two-channel decomposition does not mean every brand should target both channels equally. The audit returns the brand's current channel position; the brand-strategy prescription depends on the brand's positioning and the cell's IL construction. A celebrity-DTC brand may correctly concentrate in the cultural channel; a heritage prestige brand may correctly concentrate in the canonical channel; the diagnostic is the audit surface, not the prescription. Conflating audit with prescription would reintroduce the single-channel framing the two-channel decomposition was designed to escape. The audit's contribution is structural: it surfaces the channel distinction so the brand-strategy team can decide whether the brand's current position matches the brand's intent, and act on the gap directly rather than negotiating which intervention class applies as a separate question.

What it means. Identity Load — operationalized through the substrate's panel design — empirically predicts the AI channel where a brand will surface, with confidence-interval rigor. A brand's cell position carries a predictive prior: high-IL cells expect cultural-channel lead; medium-IL cells expect canonical-channel lead; low-IL cells lie between under monotonic-gradient discipline. v0.21 cosmetics is the program's first CONFIRMED moderator verdict at any layer (Cell B = +7.13 with CI [+4.75, +10.25]; Cell A = −3.13 with CI [−6.00, −0.25]; Cell C = +1.00 with CI [−0.50, +3.25]; all in IL-predicted regions). Brand managers auditing brands can compare a brand's measured channel asymmetry against the cell's expected asymmetry directly — deviation from expectation is itself diagnostic information. **What it doesn't mean.** Identity Load is operationalized through the substrate's panel design, not through independently-measured consumer perceptions of the brand. The moderator operates on the panel-design construction, not on a separately-measured consumer-side construct of brand identity. The construct's mechanism is the IL-gradient property of the substrate-as-measured, and the brand-strategy implication is bounded to substrates where the panel-design IL classification meaningfully tracks the brand-of-interest's category positioning. v0.20 skincare's PARTIAL on the same test surfaced substrate-specific architecture — the cell where canonical authority concentrates is substrate-dependent, not panel-design-invariant. Clinical and dermatologist-anchored authority in skincare lives in a different cell position than editorial-prestige authority in cosmetics. The moderator holds for the substrate; the cell-architecture assumption requires substrate-aware interpretation.

What it means. Methodology v1.6 closed three measurement gaps the prior framework versions could not address: substrate-level Recognition pre-screen (so maximally-coded categories like cosmetics don't collapse the verdict matrix); independent moderator pathway (so Identity-Load asymmetry is evaluable separately from the four-regime test); Phantom Brand Persistence Phase B extension (so off-panel brand presence is a measured component rather than a descriptive side observation). Together the three form a coordinated audit pathway for substrates where prior framework versions would have collapsed the verdict — Increment 1 preserves the substrate's evidentiary value, and Increments 2 and 3 carry the discrimination signal that Recognition can't supply when it's exhausted at ceiling. **What it doesn't mean.** SHIPPED v1.6 does not mean the construct is methodology-complete. v1.7 (if it ships) would extend the protocol further — the program's methodology arc is ongoing, not closed. The v1.6 increments are bounded to the retrospective scope where the v1.6 paper's published rules permit them (Increment 2 for v0.20 + v0.21; Increment 3 for v0.21 only). Cross-substrate replication of Increments 2 and 3 is v0.22+ work. AIAS™ 1.0 names the Presence component of the AIAS construct, not the full multi-component composite — and the version-number arc 1.0 → 6.0 will track the measurement surface (each new component reaching falsifiable measurement) rather than architectural ambition.

About the Third System

The Third System describes a shift in how consumers make decisions: instead of browsing and comparing options directly, many now rely on AI systems to generate recommendations. In this environment, brands compete not only for human attention but for inclusion within AI-generated outputs. This creates a new competitive layer where visibility is determined by how AI systems interpret and surface brands.

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Methodology

This report presents findings from Third System's AI Availability research program. The AI Availability Score (AIAS) is an early-stage measurement framework introduced in Tri-System Brand Growth (Gonzalez Castro, 2026). Findings should be used directionally. Construct validity — the correlation between AI Availability and consumer behavior — has not yet been established and is the subject of Phase 3 of the research program. Methodology is published in full and versioned at thirdsystem.ai/methodology (current: v1.6). Registry construction is curated and reflects the framework's view of each category's competitive set; revisions are documented in the methodology log.

Citation

Gonzalez Castro, P. U. (2026). ***AI Availability as a Third Measurable Layer of Brand Availability: Five-Substrate Empirical Anchoring of the AIAS™ Presence Measurement Protocol*** (AIAS 1.0). Third System. thirdsystem.ai/aias-1-0 (academic companion: SSRN 6817841)

Underlying datasets

AIAS 1.0 synthesis data (cross-phase source-of-truth, locked at aias-1-0-data-locked): osf.io/ec6wh/aias_1_0/aias_1_0_synthesis_data.json

Phase-level acquisitions, scoring code, and verdict matrices (v0.16 through v0.21): osf.io/ec6wh/v16/ through osf.io/ec6wh/v21/ (six per-phase deposit trees)

Methodology papers v1.2 through v1.6 (pre-registrations, retrospective scoring outputs, increment specifications): osf.io/ec6wh/methodology/v1_2/ through osf.io/ec6wh/methodology/v1_6/

AIAS 1.0 charts (academic synthesis paper figures + brand-format report upgrades): osf.io/ec6wh/aias_1_0/figures/

Pre-registration lock artifacts (outline, data, charts, paper): git tags aias-1-0-outline-locked, aias-1-0-data-locked, aias-1-0-charts-locked, aias-1-0-paper-locked in the repository

Methodology log: AIAS 1.0 synthesis paper locked at git tags aias-1-0-outline-locked, aias-1-0-data-locked, aias-1-0-charts-locked, and aias-1-0-paper-locked. Academic companion deposited at SSRN 6817841 (May 2026). Methodology version: v1.6 (SSRN 6816340). This brand-format report ships two weeks behind the SSRN deposit per the program's two-register discipline.

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